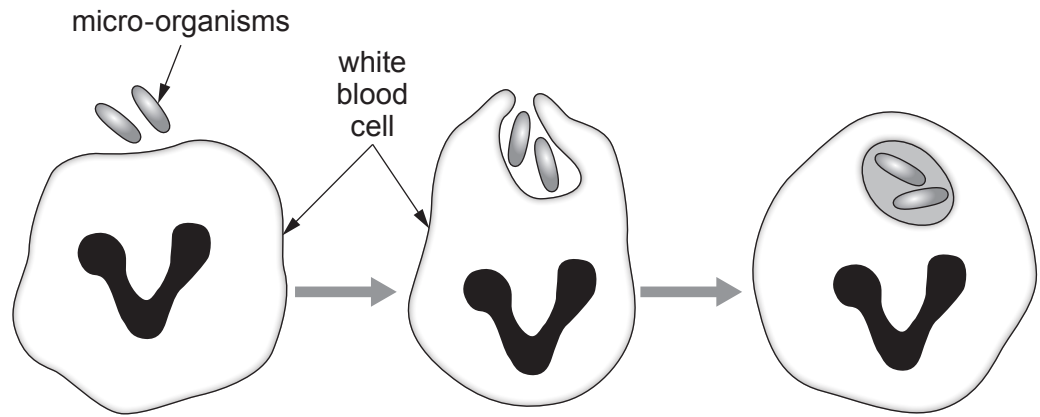


5. **Image 5.1** shows a white blood cell which is ingesting micro-organisms in the bloodstream of a person.

Image 5.1



(a) (i) State the type of white blood cell shown. [1]

.....

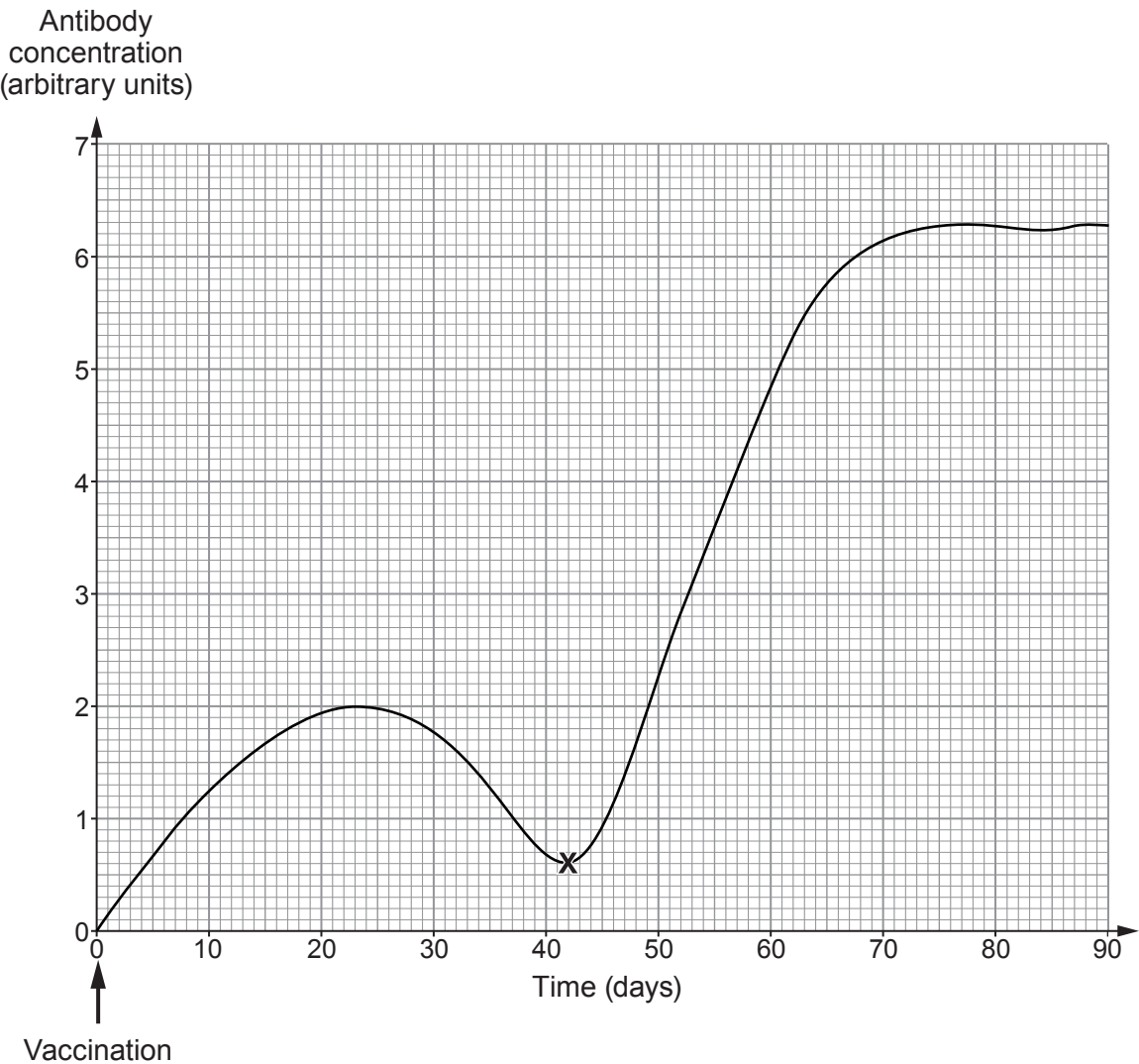
(ii) Suggest what will happen to the micro-organisms **after** they have been ingested by the white blood cell. [1]

.....
.....



- (b) A person was vaccinated against an infectious disease caused by a virus. **Graph 5.2** shows the concentration of antibodies in the blood of the person over the next 90 days.

Graph 5.2



Examiner
only

(i) Explain the increase in antibody concentration over the first 20 days. [3]

.....

.....

.....

.....

.....

.....

(ii) Give **two** possible reasons that could account for the increase in antibody concentration from point **X** on **Graph 5.2**. [2]

1.

.....

2.

.....

(iii) State the name of the cell that is responsible for the very rapid antibody production between days 45 and 65. [1]

.....

(iv) State the scientific term which could be used to describe this person in relation to the virus after day 65. [1]

.....

